Locality at $n = 9$ from the cyclic 1-zeros

via the block-rule Laurent cascade

(satellite for the $\mathrm{Tr}(\phi^3)$ locality program)

Jonathan Valenzuela

Abstract

We prove locality at $n = 9$ from the cyclic 1-zero conditions alone: every non-local coefficient of the most general rational ansatz with mass dimension $-2(n-3)$ and at most simple planar poles vanishes once the 9 cyclic hidden-zero constraints are imposed. The proof combines Step-1 (Layer-0) elimination with a depth-1 Laurent cascade. Step-1 kills 904,752 of the 905,763 non-triangulation size-6 multisets directly. The remaining 1,011 multisets fall into 113 cyclic-orbit equivalence classes, each non-local (every representative is either a crossing configuration or a double-pole). Of these: 23 orbits admit independent single-orbit depth-1 cascade recipes; the other 90 orbits form a coupled cluster $\mathcal{C}_{90}$ whose depth-1 fingerprint matrix has 506 rows and column rank 90 on cluster columns, forcing the cluster's 90 coefficients to vanish *jointly*. A complete per-orbit audit confirming all 113 kill mechanisms is recorded in the consolidated artifact at `computations/step4_laurent_block_analysis/n9/outputs/n9_locality_status.md`. This is the first polygon size at which the cluster mechanism is required: at $n = 7, 8$ every Step-1 survivor admits a single-orbit cascade. The block-rule mechanism generalises the $n = 6$ perfect-matching block (rank 3 on 4 orbits, residual 1, closed by a depth-2 anchor) to a setting where the equation count overdetermines the cluster (residual 0, no anchor needed). Unitarity (equality of triangulation coefficients to a single common scalar) is independent of this paper and follows from the d-subset / homogeneity-rescaling argument recorded in [6, 7, 4]; locality is the load-bearing half of the $\mathrm{Tr}(\phi^3)$ uniqueness theorem and is what this paper closes at $n = 9$.

1 Background: the three layers of kill mechanism

We adopt the conventions of the root README. The cyclic hidden-zero zone $\mathcal{Z}_{r,r+2}$ has special chord $X_{r,r+2} =: X_*$, bare chord $X_{r+1,r-1} = -X_*$, and substitutions $X_{r+1,k} = X_{r,k} - X_*$ for k in cyclic range. The general ansatz is $B = \sum_M a_M / \prod_{c \in M} X_c$ over size-$(n-3)$ chord multisets M.

Definition 1 (Three layers of kill mechanism)**.**

- **Step-**1 **(Layer-**0**)**: at zone $\mathcal{Z}_{r,r+2}$, send $X_* \to \infty$. The leading-order coefficient identity forces $a_M = 0$ when $M \in \mathcal{K}_{r,r+2}$, i.e. when (K1) M contains neither special nor bare and (K2) M contains no (companion, substitute) pair. Holds for M killable at *any* cyclic zone.
- **Step-**2 **(equate)**: keep X_* in the surviving variables and exploit $X_{r+1,r-1} = -X_*$. This produces equality identities $a_{M \cup \{X_{r+1,r-1}\}} = -a_{M \cup \{X_*\}}$ between coefficients differing by a bare/special swap. Step-2 is the unitarity engine; locality does *not* require it.
- **Step-**3 **(Laurent cascade)**: at zone $\mathcal{Z}_{r,r+2}$, expand

$$\frac{1}{X_{r+1,k}} = \frac{1}{X_{r,k} - X_*} = -\frac{1}{X_*} - \frac{X_{r,k}}{X_*^2} - \frac{X_{r,k}^2}{X_*^3} - \cdots$$

and extract the coefficient of a free-variable monomial (the *fingerprint*) at order $X_*^{-(\ell_r(M)+1)}$. This is a linear identity among coefficients a_M of multisets contributing the same fingerprint.

Definition 2 (Cyclic orbit; cousin)**.** $\mathbb{Z}_n$ acts on size-$(n-3)$ multisets by vertex shift; an *orbit* is a $\mathbb{Z}_n$-equivalence class. Given a Step-3 fingerprint equation $\sum_C \alpha_C\, a_C = 0$ at $(\mathcal{Z}, k, \mathcal{F})$, the multisets $\{C : \alpha_C \neq 0\}$ are the *cousins* of any M in the equation; their orbit-classes are the *cousin orbits*.

2 Single-orbit cascade and block-rule cascade

Definition 3 (Single-orbit cascade)**.** Orbit $\mathcal{O}$ admits a *single-orbit depth*-1 *cascade* recipe iff there is a triple $(\mathcal{Z}, k, \mathcal{F})$ with $k = \ell_{\mathcal{Z}}(M) + 1$ for some $M \in \mathcal{O}$ such that: (i) the fingerprint equation has a_M with nonzero scalar; and (ii) every cousin of M in this equation is Step-1-killable at some cyclic zone.

When (i)+(ii) hold, substituting cousin-zeros into the fingerprint equation collapses it to $\alpha_M\, a_M = 0$, so $a_M = 0$. Cyclic equivariance rotates the recipe onto every member of $\mathcal{O}$.

The mechanism that closes $n = 7, 8$ is: *every Step*-1 *survivor orbit admits a single-orbit recipe.* The $n = 9$ data falsifies this for 4 orbits, and a further 14 orbits time out under a 600-second search cap; the $4 + 14$ all lie in a single coupled cluster, and are killed jointly by the block-rule below rather than by individual cascades.

Definition 4 (Coupling graph; cluster)**.** The *coupling graph* G_n has vertex set = Step-1 survivor orbits at n. An edge $\mathcal{O} \sim \mathcal{O}'$ exists iff there is a Step-3 fingerprint equation with $\mathrm{orbit}(M) = \mathcal{O}$ and $\mathrm{orbit}(C) = \mathcal{O}'$ for some non-local cousin C. A *cluster* is a connected component of G_n.

Definition 5 (Block-rule cascade)**.** Let $\mathcal{C} = \{\mathcal{O}_1, \ldots, \mathcal{O}_m\}$ be a cluster. Enumerate all admissible depth-1 fingerprint recipes for orbits in $\mathcal{C}$ and assemble a matrix $M_{\mathrm{FP}}^{\mathcal{C}}$ with rows = fingerprint equations and columns = cluster orbits (*cluster columns*) plus external orbits (Step-1 survivors not in $\mathcal{C}$, plus triangulation orbits appearing as cousins). The cluster is *block-rule killable at depth*-1 iff the cluster sub-matrix has full column rank $m = |\mathcal{C}|$, given that each non-cluster non-local survivor column is set to 0 via its own single-orbit cascade.

Proposition 6 ($n = 6$ baseline: 4-orbit perfect-matching cluster)**.** *At $n = 6$, after Step*-1 *exhausts, four perfect-matching orbits survive: $M_A = \{X_{13}, X_{25}, X_{46}\}$, $M_B = \{X_{14}, X_{25}, X_{36}\}$, $M_C = \{X_{14}, X_{26}, X_{35}\}$, $M_D = \{X_{15}, X_{24}, X_{36}\}$. The depth*-1 *fingerprint matrix $M_{\mathrm{FP}}^{\mathcal{C}}$ over the six cyclic zones is a 6×4 block of rank* 3*, with nullspace spanned by $(1, -1, 1, 1)$. The residual $r = 1$ is closed by a depth*-2 *anchor: at $\mathcal{Z}_{1,3}$, the order-X_{13}^1 coefficient of $1/(X_{14}X_{15}^2X_{36})$ uniquely isolates a_{M_B} after Step*-1 *cousins die at $\mathcal{Z}_{4,6}$. Together, $a_{M_A} = a_{M_B} = a_{M_C} = a_{M_D} = 0$.*

The proof is in `paper/step2 Laurent series for hard kills/cascade_n7.tex` (precursor calculation) and the companion `fingerprint_lemma_5pt_6pt.tex`. The verification script `verify_6pt_anchor_kill.py` extracts the anchor symbolically.

3 The $n = 9$ result

Step-1 enumeration at $n = 9$ yields 1,011 non-triangulation survivors organised into 113 cyclic orbits (one of size 3, $\mathbb{Z}_3$-stabilised; 112 of size 9, free orbits). Every one of the 113 orbit representatives is a non-local size-6 multiset: 98 are crossing configurations and 15 are double-pole configurations[1].

[1] Verified by the consolidated audit in [1].

3.1 Single-orbit pass

The script `computations/step3_laurent/cascade_n9/cascade_kill_n9.py` attempts a single-orbit depth-1 recipe for each of the 113 representatives, with a 600-second timeout per orbit, then a diagnostic re-run with extended budget on the timeouts. The final disposition:

Outcome		count
single-orbit recipe found (after re-run)		95
search timeout (recipe still undetermined)		14
search exhausted, no recipe found	4	$\{\mathcal{O}_{22}, \mathcal{O}_{46}, \mathcal{O}_{88}, \mathcal{O}_{108}\}$

The 4 exhausted-search failures are diagnosed in `.../n9/diagnostics/diagnose_orbit_22.md`; they are not artifacts of bounded search depth or pruning. For example, $\mathcal{O}_{22}$ has representative $M_{22} = \{X_{1,3}, X_{1,4}, X_{1,7}, X_{1,8}, X_{4,6}, X_{5,7}\}$ and at every depth $1 \leq k \leq 4$ each fingerprint equation contains at least one cousin that is itself a Step-1 survivor; specifically $\mathcal{O}_{22}$ couples directly to $\mathcal{O}_{88}, \mathcal{O}_{96}, \mathcal{O}_{102}$ at the $\mathcal{Z}_{3,5}$ depth-1 fingerprint $X_{3,6}/(X_{1,3}X_{1,7}X_{1,8}X_{5,7})$. Single-orbit cascade cannot kill $\mathcal{O}_{22}$ in isolation; the four failures require collective treatment.

3.2 Cluster BFS and matrix

A breadth-first traversal of the coupling graph G_9 starting from $\mathcal{O}_{22}$ absorbs 90 orbits (all 4 failures and all 14 timeouts among them). The remaining 23 orbits are isolated nodes of G_9, each killed by its own verified single-orbit recipe.

For the 90-orbit cluster $\mathcal{C}_{90}$, the depth-1 cascade matrix tabulates 506 fingerprint equations against 143 columns (90 cluster orbits + 32 triangulation orbits appearing as cousins + 21 non-cluster non-local survivor orbits). Working over $\mathbb{Q}$:

quantity	value
cluster size $\lvert\mathcal{C}_{90}\rvert$	90
depth-1 fingerprint equations	506
external columns (cluster-touched triangulation + non-cluster non-local)	$32 + 21 = 53$
$\mathrm{rank}(M_{\mathrm{FP}}^{\mathcal{C}_{90}})$ on cluster columns	**90**
cluster residual $r = \lvert\mathcal{C}_{90}\rvert - \mathrm{rank}$	**0**

Theorem 7 ($n = 9$ block-rule kill)**.** *The depth-1 fingerprint sub-matrix on the 90 cluster columns of $M_{\mathrm{FP}}^{\mathcal{C}_{90}}$ has full column rank 90. Consequently, given that the 21 non-cluster non-local survivor coefficients vanish (each by its own single-orbit cascade), every one of the 90 cluster non-local coefficients vanishes:*

$$a_{\mathcal{O}_i} = 0 \qquad \forall\, \mathcal{O}_i \in \mathcal{C}_{90}.$$

Proof. The rank computation is symbolic over $\mathbb{Q}$ via `sympy`; the trace is in `cluster_partial.json`. With non-cluster non-local survivor columns set to 0, the cluster matrix gives a homogeneous linear system on the 90 cluster columns plus 32 triangulation columns. Full column rank on the cluster columns forces every cluster coefficient to be zero: the assignment "triangulations equal a common scalar c, non-cluster non-locals $= 0$, cluster $= 0$" satisfies the system (because A_n^{tree} does, putting this vector in the nullspace), and full column rank on cluster columns means this is the unique solution for cluster variables under any consistent triangulation assignment. □

3.3 Locality at $n = 9$, unconditional

Theorem 8 ($n = 9$ locality)**.** *Suppose* $B|_{\mathcal{Z}_r} = 0$ *for all nine cyclic zones at* $n = 9$*. Then* $a_M = 0$ *for every non-triangulation multiset* M *of size* 6*.*

Proof. Partition the 905,763 non-triangulation multisets at $n = 9$ by Step-1 killability:

(i) **Step-1 killed.** 904,752 multisets are Step-1 killable at some zone, so their coefficients vanish by Definition 1.

(ii) **Non-cluster Step-1 survivors.** The remaining 1,011 multisets form 113 cyclic orbits; 23 of these are isolated nodes of G_9 and admit single-orbit depth-1 cascade recipes (§3.1, audited in [1]). Each recipe forces its orbit's coefficient to zero.

(iii) **Cluster Step-1 survivors.** The remaining 90 orbits form $\mathcal{C}_{90}$. By Theorem 7, all 90 cluster non-local coefficients vanish.

Cases (i)–(iii) exhaust all non-triangulation multisets. □

Corollary 9 ($n = 9$ amplitude reconstruction)**.** *Combined with the d-subset / homogeneity-rescaling unitarity argument of [6, 7, 4], which equates all triangulation coefficients to a common scalar c,* $B = c \cdot A_9^{\text{tree}}$*.*

The unitarity input is logically independent of this paper. Locality (Theorem 8) closes the load-bearing half of the $\mathrm{Tr}(\phi^3)$ uniqueness claim at $n = 9$; the d-subset argument closes the unitarity half.

3.4 Comparison with $n = 6$

	$n = 6$ block	$n = 9$ block
cluster size m	4	90
fingerprint equations	6	506
matrix rank on cluster columns	3	90
residual r	1	0
anchor required?	yes (depth-2 at X_*^1)	no

The $n = 6$ cluster is barely large enough to determine itself, leaving a one-dimensional residual that requires an anchor kill at the next Laurent order. The $n = 9$ cluster has a roughly 5× ratio of equations to orbits and is grossly overdetermined: the depth-1 system alone closes the cluster. The $n = 6$ "anchor required" case is plausibly a small-n artifact: at higher n the fingerprint-equation count grows polynomially with cluster size, and the cluster matrix becomes generically full-rank on its cluster columns.

4 Geometric interpretation

The cascade mechanism is the algebraic shadow of a geometric operation. At zone $\mathcal{Z}_{r,r+2}$, the substitution $X_{r+1,k} = X_{r,k} - X_*$ encodes the *ear-collapse* that identifies vertex $r+1$ with the rest of

the polygon. The bare relation $X_{r+1,r-1} = -X_*$ is the 4-*gon kinematic identity*: the only chord relation at $n = 4$ is $X_{1,3} = -X_{2,4}$, and every higher-n collapse uses this identity at one local cyclic position.

The Laurent expansion in $1/X_*$ records the *kinematic memory* of the collapse: each non-bare substitute $X_{r+1,k}$ remembers its companion $X_{r,k}$ at order $X_*^{-(j+1)}$ (coefficient $X_{r,k}^j$). Depth-d probes the d-th layer of this memory.

A non-local multiset killable by a depth-d cascade has d chord-pairs whose crossing structure interacts with the relevant ear-collapse. A multiset *not* killable by any single-zone cascade—the cluster case—has crossings coupled across multiple cyclic orbits: configurations reachable from each other by a single ear-collapse move. The cluster is the connected component of the associahedron *face graph* restricted to ear-collapse moves carried by the chosen non-local cousins.

The fact that the depth-1 system on a cluster has rank $\geq |\mathcal{C}| - 1$ is, geometrically, the statement that a connected face of the associahedron has only finitely many "free directions" left after the cyclic-1-zero constraints are imposed. The anchor kill at $n = 6$ is the 1-form needed to close the last degree of freedom on a small face; at $n = 9$ the larger face is rigidified by sheer constraint count.

5 Refined all-n locality conjecture

The $n = 9$ data sharpens the all-n statement.

Conjecture 10 (Block-rule cascade locality)**.** *For every $n \geq 7$ and every cluster $\mathcal{C}$ of Step-1 survivor orbits at n:*

(1) *the cluster $\mathcal{C}$ is finite;*

(2) *there exists $r = r(\mathcal{C}) \geq 0$ with $\operatorname{rank}(M_{\mathrm{FP}}^{\mathcal{C}}) = |\mathcal{C}| - r$ on cluster columns;*

(3) *if $r \geq 1$, the residual r-dimensional space is closed by r higher-Laurent-order anchor equations.*

Equivalently: every non-local coefficient of an n-point ansatz satisfying the n cyclic 1-zero conditions is forced to vanish by Step-1 together with the depth-1 block-rule cascade and a finite collection of higher-Laurent anchors.

Corollary 11 (All-n locality, conditional)**.** *Conjecture 10, together with the Step-1 exhaustivity recorded in [2] and the triangulation-survives-Step-1 lemma of [3], implies $a_M = 0$ for every non-triangulation M at every n.*

The conjecture is verified at $n = 6, 7, 8, 9$:

- $n = 6$: $|\mathcal{C}| = 4$, $r = 1$, one depth-2 anchor closes the residual (Proposition 6).
- $n = 7, 8$: every cluster has $|\mathcal{C}| = 1$ (single-orbit case), $r = 0$.
- $n = 9$: $|\mathcal{C}_{90}| = 90$, $r = 0$ (Theorem 7); plus 23 singleton clusters, each $r = 0$.

The pattern across these data points is consistent with the prediction $r \to 0$ as n grows: the fingerprint-equation count overdetermines large clusters, and only small-cluster low-n cases require anchors.

6 What remains for the all-n locality theorem

The proof architecture is fully laid out. The remaining mathematical obligations for the all-n locality theorem are:

1. **Cluster finiteness.** Every cluster $\mathcal{C}$ at every n is finite. Likely follows from finiteness of size-$(n-3)$ chord multisets, but should be recorded as a lemma.
2. **Cluster rank lower bound.** Show $\operatorname{rank}(M^{\mathcal{C}}_{\mathrm{FP}}) \geq |\mathcal{C}| - 1$ on cluster columns for all n. This is the core combinatorial-geometric content. A generating-function argument counting fingerprint equations vs cluster orbits is plausible; an associahedron-face rigidity argument is the structural counterpart.
3. **Anchor existence when $r = 1$.** When the residual is 1-dimensional (the $n = 6$-style case), exhibit a higher-Laurent-order anchor that closes it. Conjectured to exist generically.

These are the only obligations for locality at all n. Unitarity (equality of triangulation coefficients) is independent and is handled by the d-subset / homogeneity-rescaling argument; see [6, 7, 4].

Files and reproducibility

- `computations/step1_layer0_kill/kill_enumeration/` — Step-1 enumeration through $n = 9$.
- `computations/step3_laurent/cascade_n9/cascade_kill_n9.py` — the single-orbit cascade engine.
- `computations/step4_laurent_block_analysis/n9/scripts/` — the cluster-BFS, matrix-rank, and anchor-search scripts.
- `computations/step4_laurent_block_analysis/n9/outputs/` — `cluster_partial.json` (the rank-90 matrix trace), `cluster_partial.md` (human-readable summary), `results_cascade_n9_reps.json` (per-orbit cascade outcomes), `n9_locality_status.md` (**the consolidated per-orbit locality audit**).
- `computations/step4_laurent_block_analysis/n9/diagnostics/` — per-anomaly walkthroughs for orbits 22, 25, 28, 34, 35, 46, 88, 108.

References

[1] *Consolidated $n = 9$ locality status.* This repository, `computations/step4_laurent_block_analysis/n9/outputs/ n9_locality_status.md`.

[2] *Step-1 kill statistics and asymptotic locality.* Companion satellite, this repository, `paper/step1 Kill technique and statistics/step-one-kill-statistics/`.

[3] *Triangulations are not Step-1 killable.* Companion satellite, this repository, `paper/step1 Kill technique and statistics/step-one-doesnt-kill-triangulations/`.

[4] *Proof of Rodina's claim $B = 0 \Rightarrow B_i = 0$.* Companion note, this repository, `paper/Details missing in Hidden zero -> unitarity Rodina proof/Proof of Rodina claim B=0->B`

[5] *The Laurent-tail kill at $n = 7$: seven layer-0 survivors and how they die.* Companion satellite, this repository, `paper/step2 Laurent series for hard kills/`.

[6] Handwritten note #11, *finale-d-subset-induction*, this repository, `notes/`.

[7] Handwritten note #18, *d-subset rigorous proof*, this repository, `notes/`.

[8] L. Rodina, *Hidden zeros are equivalent to enhanced ultraviolet scaling and lead to unique amplitudes in* $\mathrm{Tr}(\phi^3)$ *theory.* arXiv:2406.04234, 2024.

[9] N. Arkani-Hamed, Q. Cao, J. Dong, C. Figueiredo, S. He, *Hidden zeros for particle/string amplitudes and the unity of colored scalars, pions, and gluons.* arXiv:2312.16282, 2023.